

Multimedia Company Security Incident

Phishing Attack & Data Integrity Breach

Overview

Responded to a cybersecurity incident involving a phishing attack that led to credential compromise, unauthorized database access, and manipulation of customer data.

Incident Summary

- Intern credentials were compromised via phishing
- Unauthorized access to internal network and customer database
- Customer records were accessed, deleted, and altered
- Legitimate user locked out while attacker remained active

Objective

Using NIST Framework, provide a clear and structured overview of a cybersecurity incident, including its root cause, impact on systems and data, response actions taken, and the controls implemented to prevent recurrence and reduce future risk.

Key Risks Identified

- Lack of MFA which enabled credential compromise
- Limited phishing awareness
- Insufficient detection capabilities

Impact (Medium-to-High Severity)

- **Confidentiality:** Customer data exposed
- **Integrity:** Data altered and deleted
- **Availability:** Temporary disruption during recovery

Response Summary

The compromised account was disabled and the incident escalated after confirming unauthorized access and data integrity issues from phishing. Customer notification and law enforcement escalation were initiated as appropriate. The database was restored from backup, with missing data re-entered. To prevent recurrence, MFA, stronger authentication, monitoring, and employee training were implemented. Prompt incident reporting and existing backup procedures enabled rapid response and recovery.

Multimedia Company Incident Report

Summary	This morning, an intern reported to the IT department that she was unable to log in to her internal network account. Access logs indicate that her account has been actively accessing records in the customer database, even though she is locked out of that account. The intern received an email earlier in the morning asking her to go to an external website and log in with her internal network credentials to retrieve a message. This method was likely used by a malicious actor to gain access to our network and customer database. Several employees have noticed that some customer records are either missing or contain incorrect data. It appears customer data was both exposed to a malicious actor and manipulated as well, jeopardizing data integrity.
Identify	The incident management team audited the systems, devices, and access policies involved in the attack to identify the gaps in security. The team found that an intern's login and password were obtained by a malicious attacker and used to access data from our customer database. Upon initial review, it appears that some customer data was deleted from the database.
Protect	The team has now implemented new authentication policies to prevent future attacks: multi-factor authentication (MFA), login attempts limited to three tries, and training for all employees on how to protect login credentials. Additionally, a new protective firewall configuration and intrusion prevention system (IPS) will be implemented by next quarter.
Detect	To detect new unauthorized access attacks in the future, the team will use a firewall logging tool and an intrusion detection system (IDS) to monitor all incoming traffic from the internet. Note: An IPS will automatically block certain traffic believed to be malicious.

Respond	The team disabled the intern's network account and informed upper management of the event. This incident involves unauthorized access to internal systems through a phishing attack, resulting in both exposure and manipulation of customer data. Because this constitutes a confirmed security breach and a potential criminal act, escalation to law enforcement is appropriate. Management will inform law enforcement and other organizations as required by local laws, and they will also contact customers by mail to inform them about the data breach. To prevent future attacks, the company is providing training to interns and employees on how to protect login credentials.
Recover	The team will recover the deleted data by restoring the database from last night's full backup. Management informed staff that any customer information entered or changed this morning would not be recorded on the backup. So, they will need to re-enter that information into the database once it has been restored from last night's backup.